

HVT-Z Series Terminations

15 kV Class High Voltage Termination for Copper Tape, Wire Shield, Lead Sheath and UniShield* Cable

*UniShield is a trademark of General Cable Technologies Corporation

Product Installation Instructions

Safety Instructions

⚠ DANGER When installing electrical power system accessories, failure to follow applicable personal safety requirements and written installation instructions could result in fire or explosion and serious or fatal injuries.

⚠ DANGER To avoid risk of accidental fire or explosion when using gas torches, always check all connections for leaks before igniting the torch and follow the torch manufacturer's safety instructions.

⚠ DANGER To minimize any effect of fumes produced during installation, always provide good ventilation of confined work spaces.

⚠ DANGER As TE Connectivity (TE) has no control over field conditions which influence product installation, it is understood that the user must take this into account and apply his own experience and expertise when installing product.

⚠ DANGER Working around energized high-voltage systems may cause serious injury or death. Installation should be performed by personnel familiar with good safety practice in handling high-voltage electrical equipment. De-energize and ground all electrical systems before installing product.

⚠ DANGER Power distribution and transmission products must be properly selected for the intended application. It must be installed and serviced by competent personnel who have been trained and understand proper safety procedures. These instructions are written for such personnel and are not a substitute for adequate training and experience in safety procedures.

⚠ DANGER Read and understand the contents of these instructions before installation and follow all locally approved procedures and safety practices before installing or operating this equipment

⚠ CAUTION These instructions cannot cover all details or variations in the equipment, procedures, or processes described, nor provide directions for meeting every possible contingency during installation, operation, or maintenance. When additional information is desired to satisfy a problem not covered sufficiently for the user's purpose, please contact your TE sales representative. These instructions are not intended to supersede or replace existing safety and operating procedures.

NOTICE Upon receipt of a product, inspect it thoroughly for damage and loss of parts incurred during shipment. If damage or loss is discovered, file a claim with the carrier immediately or contact your TE representative.

Suggested Installation Equipment (not supplied with kit)

- Cable preparation tools
- TE P63 cable preparation kit or cable manufacturer approved solvent, use grit cloth and solvent if provided in kit.
- Clean, lint-free cloths
- Non-conducting abrasive cloth, 120 grit or finer
- Electrician's tape
- Connector(s) and installation tools
- TE recommended torch

Recommended TE Torches

Install heat-shrinkable cable accessories with a "clean burning" torch, i.e., a propane torch that does not deposit conductive contaminants on the product.

Clean burning torches include the TE FH-2629, FH-2649 (uses refillable propane cylinders) and FH-2618A (uses disposable cylinder).

Adjusting the Torch

Adjust regulator and torch as required to provide an overall 12- inch bushy flame. The FH-2629 will be all blue, the other torches will have a 3- to 4-inch yellow tip. Use the yellow tip for shrinking.

Regulator Pressure

FH-2618A	Full pressure
FH-2649	25 psig
FH-2629	15 psig

General Shrinking Instructions

- Apply outer 3- to 4-inch tip of the flame to heat-shrinkable material with a rapid brushing motion
- Keep flame moving to avoid scorching
- Unless otherwise instructed, start shrinking tube at center, working flame around all sides of the tube to apply uniform heat

To determine if a tube has completely recovered, look for the following, especially on the back and underside of the tube:

1. Uniform wall thickness
2. Conformance to substrate
3. No flat spots or chill marks
4. Visible sealant flow if the tube is coated

NOTICE When installing multiple tubes, make sure that the surface of the last tube is still warm before positioning and shrinking the next tube. If installed tube has cooled, re-heat the entire surface.

Customer Service

For 24 hour customer service, call 800-327-6996.

Kit Contents

- 1 Red, non-tracking tube
- 2 Red mastic strips
- 1 Yellow angle-cut mastic strip
- 1 Ground clamp spring
- 1 Solder-blocked braid
- 2 Strip of copper-backed tape
- 1 Set of Installation Instructions
- 1 Red heat-shrinkable skirt (for outdoor terminations only)

Installation Instructions

1. Product selection

Check kit selection with cable diameter dimensions in Table 1.

For three conductor terminations, refer to the installation instructions supplied in the MOD-3-HVT kit (ordered separately) to seal the phase conductors before installing the single conductor terminations

Table 1

Kit	Conductor Size 15 kV	Minimum/Maximum Insulation Diameter	Maximum Jacket Diameter
HVT-Z-151-G/SG	#4 - 1/0 AWG	0.60-0.95" (15-24mm)	1.20" (30mm)
HVT-Z-152-G/SG	2/0 - 350 kcmil	0.80-1.25" (20-32mm)	1.50" (38mm)
HVT-Z-153-G/SG	400 - 1000 kcmil	1.10-1.75" (28-44mm)	2.10" (53mm)
HVT-Z-154-G/SG	1250 - 2500 kcmil	1.60-2.45" (41-62mm)	2.75" (70mm)

2. Prepare cables

Choose the cable type (Choice 1-4) and follow the directions given.

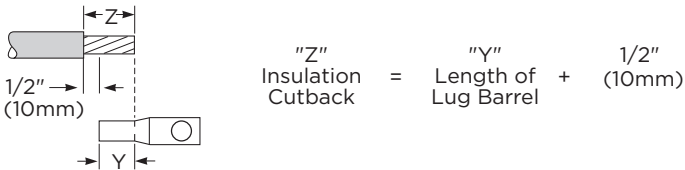
NOTICE Jacket cutback is based upon a max. lug barrel length of 3-3/4". If the lug barrel is longer, the additional length must be added to jacket cutback to maintain proper termination stress control.

Figure: Insulation Cutback (Z)

With lug: Zmax=4 1/4" unless jacket cutback is lengthened

Without lug: Z=2" (50mm)

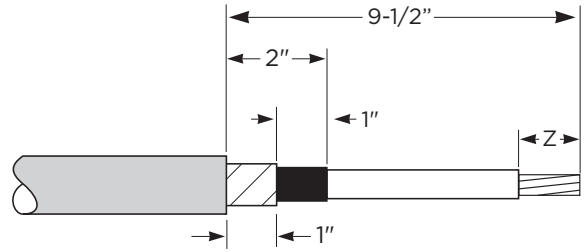
For longer Z (exposed conductor) add desired Z minus 2" to jacket cutback dimension.



Choice 1

Copper Tape Shield Cable

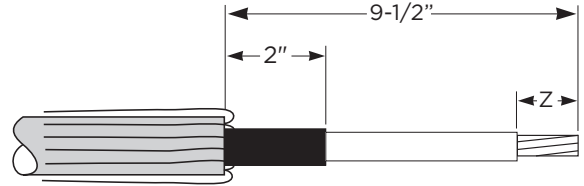
Prepare as shown, then continue with Step 3, option A.



Choice 2

Wire Shield Cable

Prepare as shown, pulling drain wires back, then continue with Step 4.

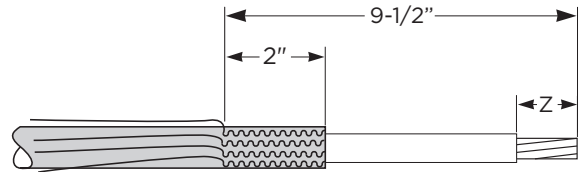


Choice 3

UniShield* Cable

*UniShield is a trademark of General Cable Technologies Corporation

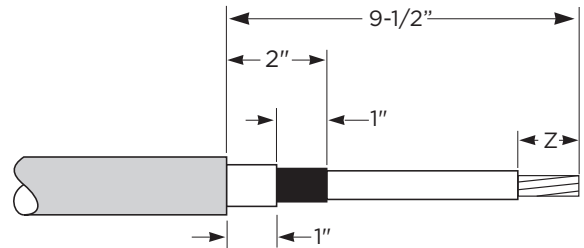
Prepare as shown, pulling drain wires back before cutting the jacket.
Continue with Step 4.



Choice 4

Lead Sheath Cable

Prepare as shown, pulling drain wires back, then continue with Step 3, option B.



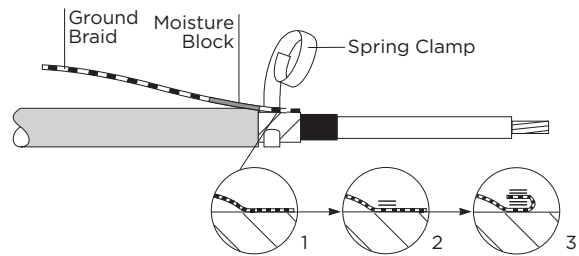
3. Install ground braid

A. For Copper Tape Shield Cable Only

(1) Lay the ground braid on the cable with the solder block aligned with the jacket cutback and the short end of the braid over the exposed copper tape shield.

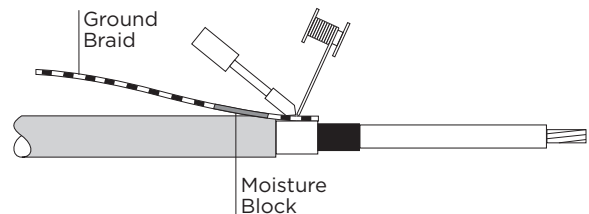
(2) Attach the braid to the shield by placing two wraps of the spring clamp over the braid.

(3) Fold the short end of the braid back over the spring clamp wraps. Continue to wrap the remaining clamp over the braid. Tighten clamp by twisting it in the direction it is wrapped and secure with the copper foil tape provided.



B. For Lead Sheath Cable Only

Lay the ground braid on the cable with the solder block aligned with the jacket cutback and the short end of the braid over the exposed lead sheath. Solder the ground braid to the lead sheath.



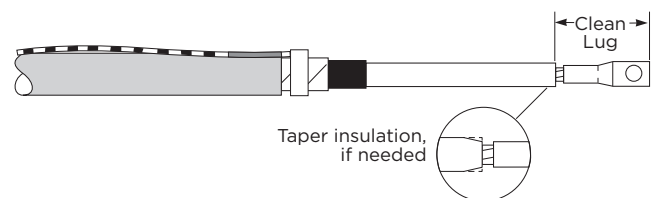
4. Install lug if needed

If no lug is used, go to Step 5.

After installation, clean and deburr lug as shown.

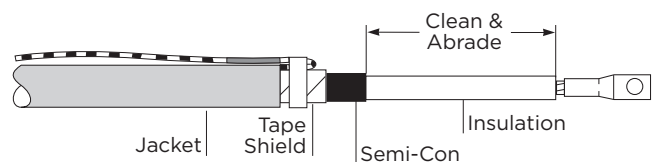
NOTICE If the step between the lug barrel and the insulation is greater than 1/8" (3mm), taper (chamfer) the insulation to meet the lug barrel. Conventional pencilling is not required.

NOTICE A copper tape cable termination is shown in this example. Any cable described earlier can be used



5. Abrade insulation; clean cable

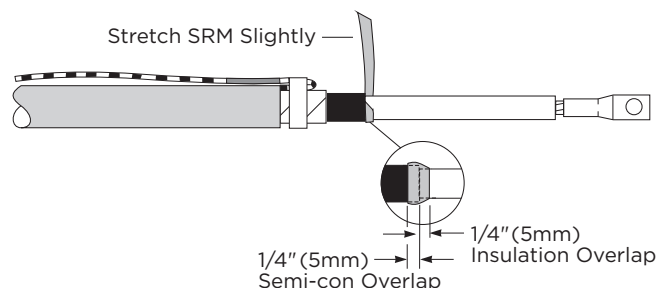
Abrade the insulation, if necessary, to remove embedded semi-con. Using an oil-free solvent, clean the cable as shown.



6. Apply stress relief material (SRM) at semi-con cutback

Remove backings from the yellow short angle-cut piece of SRM. Place tip of SRM at semi-con cutback and stretch to approximately 1/2 its original width. Tightly wrap 3 to 4 layers to fill semi-con step. Overlap semi-con and insulation as shown. Taper SRM down to meet insulation.

NOTICE If using UniShield cable, apply SRM as shown to fill conductive jacket step.



7. Complete grounding

Choose the cable type (Choice 1 or 2) and follow the directions given

Choice 1

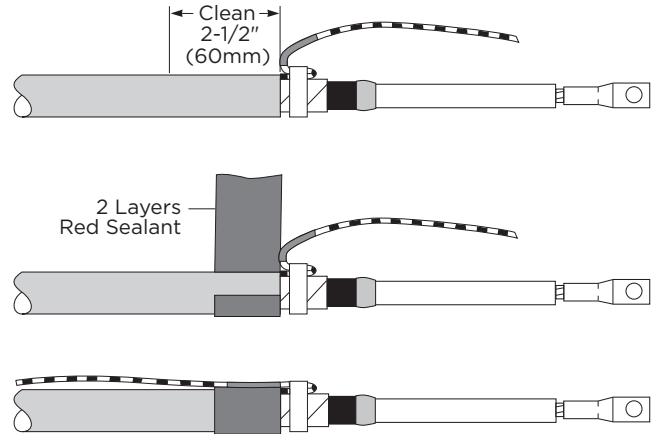
Copper Tape Shield or Lead Sheath Cable

Lift ground braid away from the cable jacket. Using an oil-free solvent, clean the jacket for 2-1/2" (60mm) below the jacket cutback point.

Remove the backing from the red sealant. Using light tension, wrap two layers of red sealant onto the jacket under the braid.

Lay the braid back over the jacket and press moisture blocked section of braid into the red sealant.

Go to Step 8.



Choice 2

Wire Shield or UniShield* Cable

*UniShield is a trademark of General Cable Technologies Corporation

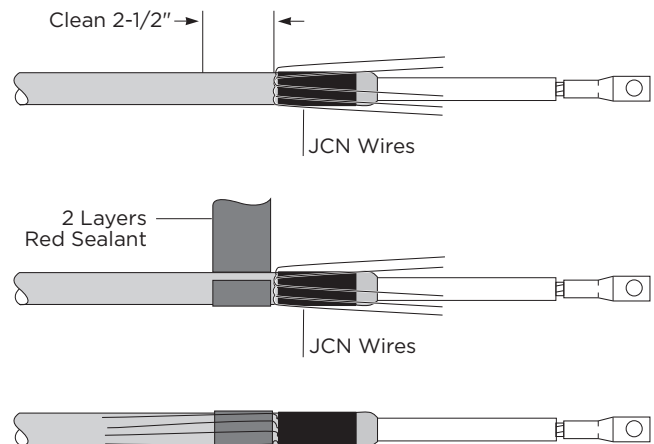
Lift the drain wires away from the cable jacket. Using an oil-free solvent, clean the jacket as shown.

Remove the backing from the red sealant. Using light tension, wrap two layers of red sealant onto the jacket under the drain wires.

Lay the drain wires evenly back over the jacket and press them into the red sealant.

Discard spring clamp, braid, and foil tape.

Go to Step 8.



8. Apply red sealant

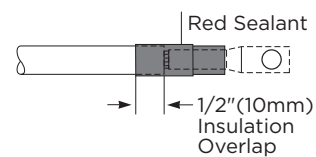
A. With Lug

Using light tension, wrap red sealant on lug barrel. Build up lug diameter to cable insulation and overlap insulation by 1/2" (10mm).

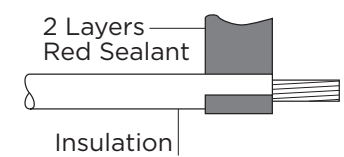
B. Without Lug

Wrap two layers of red sealant onto the cable insulation as shown.

A. With Lug



B. Without Lug

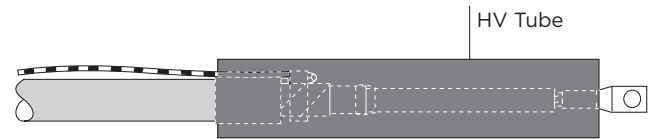


NOTICE

Sealant will not prevent water ingress between the strands.

9. Position the HV tube in place

Preheat the termination lug before placing the tube over the cable. The bottom end of the tubing should be level with the red sealant on the cable jacket.

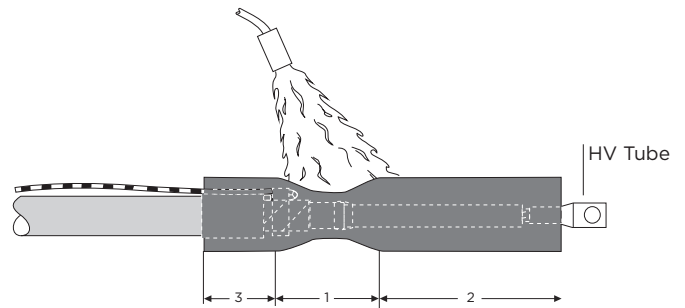


10. Shrinking HV tube

Shrink the tubing down starting at the semi-con cutback. Heat the area well but avoid scorching the surface. Continue shrinking towards the cable lug. Finally shrink down the bottom end of the tubing. The numbers in the drawing indicate the shrink sequence.

NOTICE After installation the termination must be post heated as well as the palm of the cable lug until a bead of sealant (green) appears around the top of the tubing. Allow the termination to cool before applying any mechanical strain.

NOTICE Due to the lubricating effect of the inner surface coating, there will be some longitudinal shrinkage of the tube.

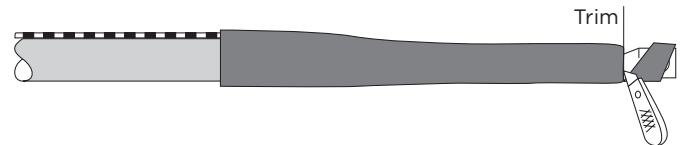


11. Trim HV tube

Allow tube to cool enough to touch before proceeding.

If necessary, trim excess tubing from the lug area as shown.

Inspect installation; reheal any flat spots or wrinkles.



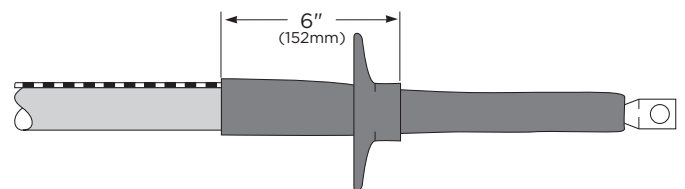
12. Indoor Termination completed

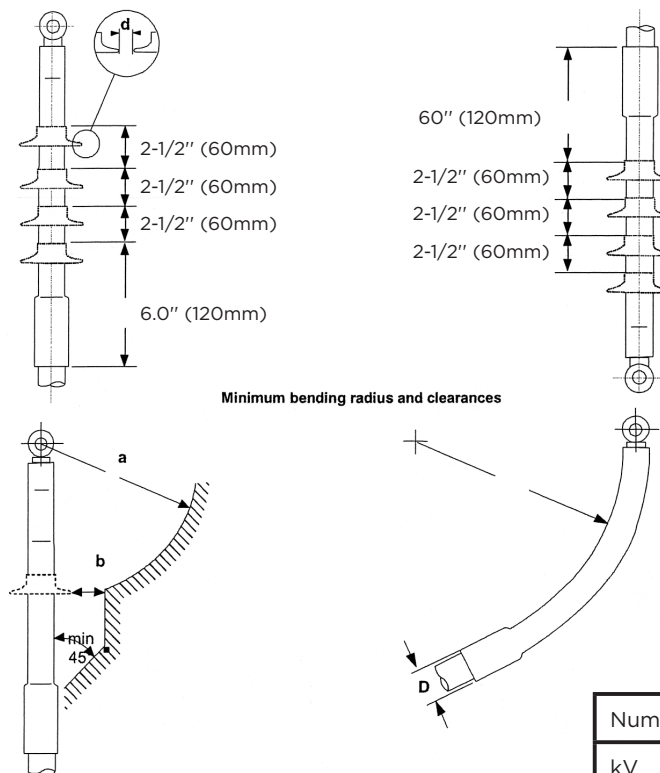


13. Outdoor Terminations only: position skirt and shrink in place.

Position the skirt as shown and shrink it in place. Hold the edge of the skirt lightly with pliers. Brush flame tip around outside of skirt collar. Only the collar will shrink.

Outdoor Termination is Complete.





Number of Skirts per phase		
kV	Indoor	Outdoor
5/8	0	0
15	0	1
25/35	0	4

Minimum Clearances	Maximum System voltage in kV			
	5/8	15	25	35
a Air Clearance	as per local specifications			
b ph/ph and ph/ground in mm	0.6" (15mm)	0.75" (20mm)	1.0" (25mm)	1.40" (35mm)
c Between skirts in mm	0.40" (10mm)	0.60" (15mm)	0.75" (20mm)	1.0" (25mm)

* Including same phase terminations.

D = Cable Jacket O.D.

Recommended Cable Bending Radius

TE terminations are as flexible as the original cables. A cable end should not be bent to a radius less than that recommended by the manufacturer.

$R = 10 \times D$ - Minimum bending radius (Consult cable manufacturer's values and check them against TE's. Select the higher of the two. Preheat cable to approximately 80 degrees before bending.)

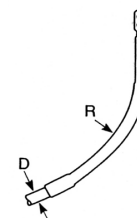


Figure 5.
Bending radius

The Information contained in these installation instructions is for use only by installers trained to make electrical power installations and is intended to describe the correct method of installation for this product. However, TE has no control over the field conditions which influence product installation. It is the user's responsibility to determine the suitability of the installation method in the user's field conditions. TE's only obligations are those in TE's standard Conditions of Sale for this product and in no case will TE be liable for any other incidental, indirect or consequential damages arising from the use or misuse of the products.

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